Breaking Universality in the Lower Order Terms in the 1-level and 2-level Density of Holomorphic Cusp Newforms

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Introduction

We study the distribution statistics of zeros of L-functions on the critical line.

- High on the critical line: universal random matrix ensemble behavior for a single L-function (Montgomery, Odlyzko, Rudnick–Sarnak).
- Near $s = 1/2$: zeros are scarce, but averaging over infinite families yields (conjectural) universal behavior.
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  - These zeros carry arithmetic information (BSD).

Question

Katz-Sarnak conjectures universal limiting behavior in the density of low lying zeros. Does the arithmetic of an L-function family affect the rate of convergence?

The family of L-functions

Fix an even weight k and let the level $N \to \infty$.

- $\mathcal{F}_N := H_k^*(N)$, the holomorphic cuspidal newforms of weight k and level N.
- Each $f \in \mathcal{F}_N$ has $L(s,f) = \sum_{n \geq 1} \lambda_f(n)\, n^{-s}$, where $\lambda_f(n)$ is the nth Hecke eigenvalue.
- We have an analytic conductor

$$R \;:=\; \frac{(k+1)(k+3)}{64\pi^2}\, N,$$

  so $\boxed{\log R \sim \log N.}$
- We assume GRH and order the zeros on the critical line as $\frac{1}{2} + i\gamma_f^{(j)}$.

The 1- and 2-level densities

For even Schwartz test functions ϕ, ϕ_1, ϕ_2, we have the 1- and 2-level densities:

$$D_1(f;\phi) := \sum_j \phi\left(\gamma_f^{(j)} \frac{\log R}{2\pi}\right), \qquad D_2(f;\phi_1,\phi_2) := \sum_{i\neq\pm j} \phi_1\left(\gamma_f^{(i)} \frac{\log R}{2\pi}\right)\phi_2\left(\gamma_f^{(j)} \frac{\log R}{2\pi}\right),$$

and their family averages

$$D_1(\mathcal{F},\phi) := \frac{\sum_{f\in\mathcal{F}} w_R(f) D_1(f,\phi)}{\sum_{f\in\mathcal{F}} w_R(f)} \qquad D_2(\mathcal{F},\phi_1,\phi_2) := \frac{\sum_{f\in\mathcal{F}} w_R(f) D_2(f,\phi_1,\phi_2)}{\sum_{f\in\mathcal{F}} w_R(f)}$$

where $w_R(f)$ is a weighting which aligns with the Petersson trace formula (Miller).

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D_1 and D_2 record "moments" of the density statistics.

The Katz–Sarnak density conjecture

Conjecture (Katz–Sarnak, 1999)

For a natural family $\mathcal{F}$ ordered by conductor, there is a symmetry type $G(\mathcal{F}) \in \{U,\ USp,\ O,\ SO(\text{even}),\ SO(\text{odd})\}$ with

$$\lim_{N\to\infty} D_n(\mathcal{F}_N;\phi) \;=\; \int \phi(x)\, W_{n,G(\mathcal{F})}(x)\, dx,$$

where $W_{n,G}$ is the n-level density of eigenvalues near 1 of Haar-random matrices in G, as the matrix size tends to infinity.

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Iwaniec–Luo–Sarnak (2000). For $\mathcal{F}_N = H_k^*(N)$, N squarefree, under GRH and $\operatorname{supp}\widehat{\phi} \subset (-2,2)$,

$$D_1(\mathcal{F}_N,\phi) \;\longrightarrow\; \widehat{\phi}(0) + \tfrac{1}{2}\phi(0).$$

The limit depends on the family *only through* $G(\mathcal{F})$: it is **universal**.

Lower order information

- Katz–Sarnak describes $R \to \infty$. Typically

$$D_1(\mathcal{F}_N, \phi) \;=\; \int \phi\, W_{1,G} \;+\; \frac{c(\mathcal{F}, \phi)}{\log R} \;+\; O\!\left(\frac{1}{\log^2 R}\right).$$

  The **lower order terms** control the *rate* of convergence, and this is where the arithmetic of the family appears.

- Convergence is in powers of $1/\log R$. At $R = 10^{10}$, $1/\log R \approx 0.043$, so at any conductor we can compute with, the lower order terms are not negligible.

Analogy: the central limit theorem

Let $X_1, X_2, \ldots$ be i.i.d. with mean 0, variance 1, $\mathbb{E}|X_1|^3 < \infty$, and $S_n := (X_1 + \cdots + X_n)/\sqrt{n}$.

- **Universal limit.** $\mathbb{P}(S_n \le x) \to \Phi(x)$, whatever the distribution of X_1.
- **Berry–Esseen.** The rate depends on the distribution:

$$\sup_x \left|\mathbb{P}(S_n \le x) - \Phi(x)\right| \;\le\; \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}}.$$

Which levels?

Our family is associated to a sequence of levels $N \to \infty$. *We care about their prime factorizations.*
Let $N \to \infty$ with a fixed factorization shape:

$$N = q_1^{a_1} \cdots q_n^{a_n}, \qquad n,\ a_1, \ldots, a_n \text{ fixed},$$

where each q_i is either a **bounded** prime (fixed), or a **growing** prime, $q_i \asymp N^{\delta_i}$ with $\delta_i > 0$.

Let Q_0 be the bounded primes dividing N, and $Q \subseteq Q_0$ those with $q \,\|\, N$.

$$\kappa_1(\mathcal{F}) := \sum_{q \in Q_0} \frac{\log q}{q} + \sum_{q \in Q} \frac{\log q}{q^2 - 1}$$

N	p or pq	$2p$	$3p$
κ_1	0	0.578	0.504

Theorem 1: the one-level density

Theorem 1

Assume GRH. Let $\mathcal{F}$, $\widetilde{\mathcal{F}}$ be two such families of weight k with levels $N \asymp \widetilde{N}$, and let $\operatorname{supp}\widehat{\phi} \subseteq [-\sigma, \sigma]$ with $\sigma < 4/3$. Then

$$\lim_{N\to\infty} \log R \left[D_1(\mathcal{F}_N, \phi) - D_1(\widetilde{\mathcal{F}}_{\widetilde{N}}, \phi) \right] \;=\; -2\,\widehat{\phi}(0)\,\big(\kappa_1(\mathcal{F}) - \kappa_1(\widetilde{\mathcal{F}})\big).$$

In particular,

$$D_1(\mathcal{F}_N, \phi) \;=\; \underbrace{\widehat{\phi}(0) + \tfrac{1}{2}\phi(0)}_{\text{Katz–Sarnak}} \;+\; \frac{\big(c_k - 2\kappa_1(\mathcal{F})\big)\,\widehat{\phi}(0)}{\log R} \;+\; O\!\left(\frac{1}{\log^3 R}\right),$$

Example: $N = p$ versus $N = 2p$

Both families are orthogonal, have the same Katz–Sarnak limit, and $R \asymp \widetilde{R}$. But

$$\kappa_1 = 0 \qquad \text{versus} \qquad \kappa_1 = \frac{\log 2}{2} + \frac{\log 2}{3} = \frac{5 \log 2}{6},$$

so Theorem 1 gives

$$\lim_{N \to \infty} \log R \left[D_1(\text{level } p) - D_1(\text{level } 2p) \right] \;=\; \frac{5 \log 2}{3} \widehat{\phi}(0) \;\approx\; 1.155 \, \widehat{\phi}(0) \neq 0.$$

Theorem 2: the two-level density

Theorem 2

Let $\mathcal{F}$, $\widetilde{\mathcal{F}}$ be families with *squarefree* levels $N \asymp \widetilde{N}$, and let $\operatorname{supp}\widehat{\phi}_1, \operatorname{supp}\widehat{\phi}_2 \subseteq [-\sigma, \sigma]$ with $\sigma < 2/3$. Then

$$\lim_{N\to\infty} \log R \left[D_2(\mathcal{F}_N, \phi_1, \phi_2) - D_2(\widetilde{\mathcal{F}}_{\widetilde{N}}, \phi_1, \phi_2) \right] \;=\; -\,\Xi_2(\phi_1, \phi_2)\,\big(\kappa_1(\mathcal{F}) - \kappa_1(\widetilde{\mathcal{F}})\big),$$

where $\Xi_2(\phi_1, \phi_2) := 4\Big(\int \phi_1 \int \phi_2 - \int \phi_1\phi_2\Big) + \phi_1(0)\int \phi_2 + \phi_2(0)\int \phi_1$.

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Example. Levels p versus $2p$: the limit is $\frac{5}{6}\log 2 \cdot \Xi_2(\phi_1, \phi_2) \neq 0$.

Proof strategy

Explicit formula. With Satake parameters $\alpha_f(p), \beta_f(p)$ and $c_m(p;f) := \alpha_f(p)^m + \beta_f(p)^m$ (Miller),

$$D_1(f;\phi) \;=\; \frac{U_{k,N}(\phi)}{\log R} \;-\; 2\sum_{p}\sum_{m\geq 1} \frac{c_m(p;f)}{p^{m/2}} \frac{\log p}{\log R}\, \widehat{\phi}\!\left(\frac{m\log p}{\log R}\right).$$

The Archimedean term $U_{k,N}(\phi)$ does not depend on the factorization of N, so it is universal.

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Prime dependence. Our computation relies heavily on when a prime does or does not divide N. For instance,

$$c_m(p;f) \;=\; \begin{cases} \lambda_f(p^m) - \lambda_f(p^{m-2}), & p \nmid N \\ \lambda_f(p)^m, & p \mid N, \end{cases} \qquad \lambda_f^2(p) \;=\; \begin{cases} 1/p, & p \,\|\, N \\ 0, & p^2 \mid N. \end{cases}$$

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Growth rates. We analyze contributions of values p and m in the sum above, and ignore sufficiently small terms ($O(\log^{-5} R)$). This is how the expression for κ_1 arises.

Future work

We may further investigate:

- Two-level density for non-squarefree N,
- Prime factors of intermediate size, e.g. $q \asymp \log N$: they contribute about $\log\log N / \log N$ to κ_1, which tends to 0, but slowly.

References

[Ber] A. C. Berry, *The accuracy of the Gaussian approximation to the sum of independent variates*, Trans. Amer. Math. Soc. **49** (1941), 122–136.

[Ess] C.-G. Esseen, *On the Liapunoff limit of error in the theory of probability*, Ark. Mat. Astr. Fys. **A28** (1942), 1–19.

[ILS] H. Iwaniec, W. Luo and P. Sarnak, *Low lying zeros of families of L-functions*, Inst. Hautes Études Sci. Publ. Math. **91** (2000), 55–131.

[KS1] N. M. Katz and P. Sarnak, *Random Matrices, Frobenius Eigenvalues, and Monodromy*, AMS Colloq. Publ. **45**, 1999.

[KS2] N. M. Katz and P. Sarnak, *Zeros of zeta functions and symmetries*, Bull. Amer. Math. Soc. **36** (1999), 1–26.

[Mil] S. J. Miller, *Lower order terms in the 1-level density for families of holomorphic cuspidal newforms*, Acta Arith. **137** (2009), 51–98.

[Mon] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24**, AMS, 1973, 181–193.

[Odl] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.

[RS] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Math. J. **81** (1996), 269–322.

Thank You

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